

Donghyun Kim

Digital Design & Verification Engineer

I develop FPGA and embedded systems with a focus on RTL design and UVM-based verification.
My work centers on digital circuit design, functional verification, and hardware/software integration.

Education

Kwangwoon University - Bachelor, Electronic Engineering

2020.03 - 2026.02

- Relevant coursework : Digital Logic Design, Computer Architecture, Semiconductor Engineering
- GPA : 4.36 / 4.5 (Major GPA : 4.5 / 4.5)
- Project :
 1. 10MHz 10bits Monotonic SAR ADC circuit design
 2. Snapshot Digital PLL circuit design
- Club : ROLAB / Trick

On-Device AI System Semiconductor Design Education

2026.03 - Current

- Project :
 1. SoC based on AXI
 2. RV32I Single-Cycle CPU
 3. UART / FIFO Design & Verification

Skills

Area	Details
RTL Design	SystemVerilog, Verilog
Verification	UVM, SystemVerilog Assertions (SVA), Functional Coverage
FPGA	Xilinx Artix-7 (Basys3), Vivado
Embedded	MicroBlaze, Embedded C
Tools	Git, GTKWave, Vivado Simulator

Featured Projects

OV7670 Camera with VGA

1. Get image from ov7670 camera and display it by VGA connect
2. SCCB is designed based on FSM to setup ov7670 camera module
3. Gray and Binary filter is added and controlled by switch on Basys3 board

OV7670 SCCB VGA Verilog

SoC based on AXI

1. AXI4-Lite based SoC implemented on the Basys3 with MicroBlaze
2. RTL design custom IP is used - Timer, UART, SPI, I2C, GPIO
3. Verification with Vitis C code

AXI Custom IP Vivado Vitis C

RV32I Pipeline CPU

1. 5 stage pipeline CPU supporting the RISC-V RV32I instruction set
2. Designed to resolve setup time violation at single cycle CPU
3. Achieved a 32.3% reduction in critical path delay compared to the single-cycle design, improving from 14.391 ns to 9.736 ns
4. Verification with assembly code - Bubble sort algorithm

Verilog RISC-V Basys3

UART & FIFO with UVM

1. RTL design of 8-N-1 UART protocol controller and 8 bit width synchronous FIFO
2. Verification with UVM

Verilog SystemVerilog UART FIFO UVM

AI Utilization

UVM Auto Generation

- Automated generation of UVM boilerplate with Python, minimizing repetitive coding unrelated to verification logic

Contact

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